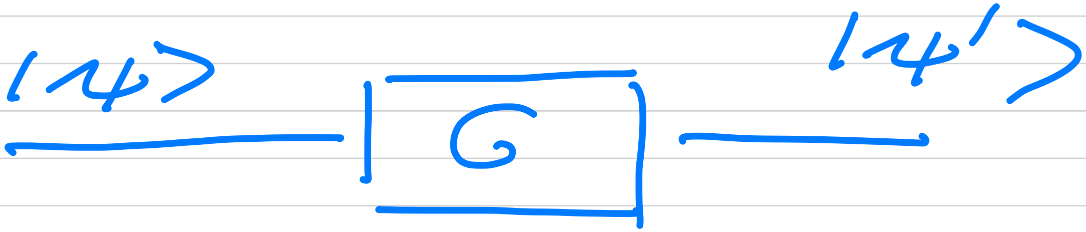
FYS5419/9419 February 11

Discussion of one and two-qubit gates.

$$|\psi'\rangle = U|\psi\rangle = G|\psi\rangle$$



$|\psi\rangle \Rightarrow$ one qubit system

computational basis

$$|0\rangle = \begin{bmatrix}1\\0\end{bmatrix} , \quad |1\rangle = \begin{bmatrix}0\\1\end{bmatrix}$$

$$\sigma_z = Z = \begin{bmatrix}1 & 0\\0 & -1\end{bmatrix}$$

$$Z|0\rangle = |0\rangle \wedge Z|1\rangle = -|1\rangle$$

$$|0\rangle \;—\; \boxed{Z} \;—\; |0\rangle$$

$$|1\rangle \;—\; \boxed{Z} \;—\; |1\rangle$$

Spin-Flip gates

$$\sigma_x = X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$X|0\rangle = |1\rangle \wedge X|1\rangle = |0\rangle$$

$$|0\rangle \longrightarrow \boxed{X} \longrightarrow |1\rangle \qquad (\text{NOT gate})$$

$$Y = \sigma_y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

$$Y|0\rangle = i\,|1\rangle$$

$$Y|1\rangle = -i\,|0\rangle$$

$$R_k = \begin{bmatrix} 1 & 0 \\ 0 & e^{i2\pi/2^k} \end{bmatrix}$$

S-gate $= \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$

$e^{i\pi/2}$

Quick reminder of state on Bloch sphere

$$|\psi\rangle = \cos(\theta/2)|0\rangle + e^{i\varphi}\sin\left(\frac{\theta}{2}\right)|1\rangle$$

Hadamard gate

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$H^T H = H H^T = \mathbb{1}$$

$$H^{-1} = H$$

$$H = \frac{1}{\sqrt{2}} \left[X + Z \right]$$

$$H X H = Z$$

$$X = HZH$$

$$HXH = \frac{1}{\sqrt{2}}[X+Z]\,X\,[X+Z]\frac{1}{\sqrt{2}}$$

$$= \frac{1}{2}\left[\underbrace{X^2}_{\mathbb{1}}\cancel{X} + \underbrace{X^2}_{\mathbb{1}}Z + Z\underbrace{X^2}_{\mathbb{1}} + \underbrace{ZXZ}_{\cancel{-X}}\right]$$

$$= \frac{1}{2}Z\cdot 2 = Z$$

$$HYH = -Y = XYX$$

Two-qubit :

control gates CU

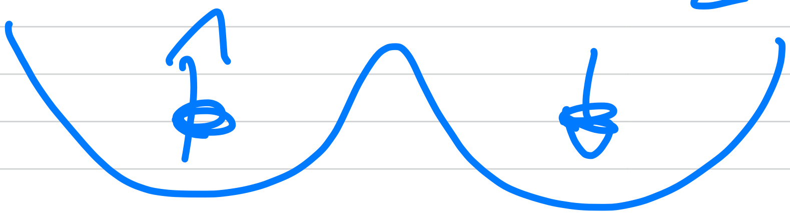
$$CU = |0\rangle\langle 0| \otimes \mathbb{I}_{2\times 2} + |1\rangle\langle 1| \otimes U_{2\times 2}$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & U_{00} & U_{01} \\ 0 & 0 & U_{10} & U_{11} \end{bmatrix}$$

$$\mathrm{CNOT} = CX \quad (CX_{01})$$

$$CX = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$(\sigma_z)_B$



A B

$$CX\,|00\rangle = \begin{bmatrix} \overset{1}{\boxed{\begin{matrix} 1 & 0 \\ 0 & 1 \end{matrix}}} & 0 \\ 0 & \overset{2}{\boxed{\begin{matrix} 0 & 1 \\ 1 & 0 \end{matrix}}} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} = |00\rangle$$

$$CX\,|10\rangle = \begin{bmatrix} \begin{matrix} 1 & 0 \\ 0 & 1 \end{matrix} & 0 \\ 0 & \begin{matrix} 0 & 1 \\ 1 & 0 \end{matrix} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

$$= |11\rangle \quad \wedge \quad CX\,|01\rangle = |01\rangle$$

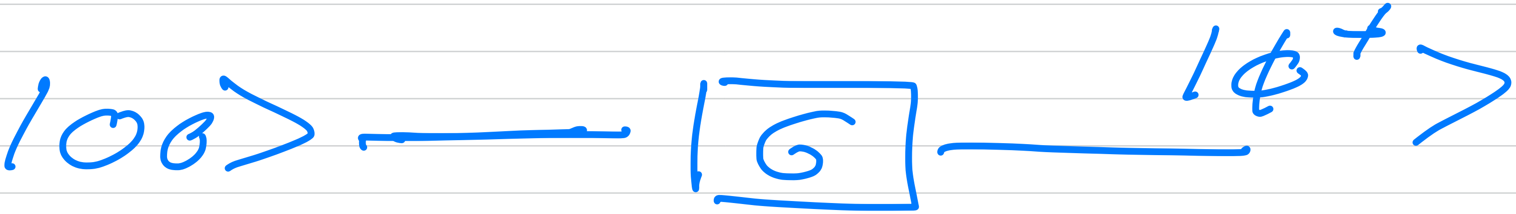
$$CX\,|11\rangle = |10\rangle$$

$$\mathrm{SWAP}\,|xy\rangle = |yx\rangle$$

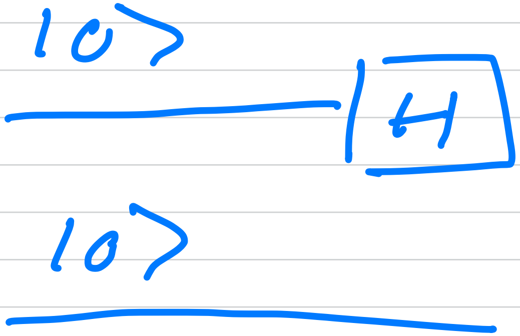
$$\mathrm{SWAP} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Bell states. How do we make them starting with $|00\rangle$

$$|\phi^+\rangle = \frac{1}{\sqrt{2}}\left(|00\rangle + |11\rangle\right)$$



Hadamard on first qubit



$$H|0\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle + |1\rangle\right)$$

$$H \otimes I\,|00\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle + |1\rangle\right) \otimes |0\rangle$$

$$= \frac{1}{\sqrt{2}}\left(|00\rangle + |10\rangle\right) = |\psi'\rangle$$

$$CX\,|\psi'\rangle = |\psi''\rangle$$

$$\frac{1}{\sqrt{2}}\begin{bmatrix}1&0&0&0\\0&1&0&0\\0&0&0&1\\0&0&1&0\end{bmatrix}\left(\begin{bmatrix}1\\0\\0\\0\end{bmatrix}+\begin{bmatrix}0\\1\\0\\0\end{bmatrix}\right)$$

$$CX\,|00\rangle = |00\rangle \wedge CX\,|10\rangle = |11\rangle$$

$$|\psi''\rangle = \frac{1}{\sqrt{2}}\left(|00\rangle + |11\rangle\right) = |\Phi^+\rangle$$

Circuit :